

AI CAMERA FOR TRACKING Suspicious Activities

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Tracking and detecting suspicious activities is one of the most demanding tasks for security personnel and systems. If a security system fails in a place like airport or bank, criminals can rob at gunpoint. Out of fear, the victims are usually unable to call the police in time for help.

But if a smart camera could detect any suspicious activity and automatically trigger an alarm to call the police, such criminal acts are likely to reduce significantly. So, let's see how to build such an alarm.

Training the camera

Create a machine learning (ML) model having datasets of normal activities, such as walking, talking, reading, and sitting. Feed the datasets of suspicious activities like fighting, boxing, pointing a gun, or any other violent movement deemed suspicious, into the ML model.

Perform the above-mentioned activities in front of the smart camera so that various movements are captured. This will help train the ML model and deploy it on Raspberry Pi as a smart artificial intelligence (AI) camera.

There are several flexible options, such as TensorFlow, Google Teachable, Edge Impulse, and Lobeo to create and train an ML model.

TABLE 1

MATERIALS REQUIRED

Component name	Quantity	Description
Raspberry Pi 4	1	Zero W or RPi 4
RPi Camera	1	Camera module
Speaker/Amplifier	1	Speaker module



Fig. 1: RPi AI camera

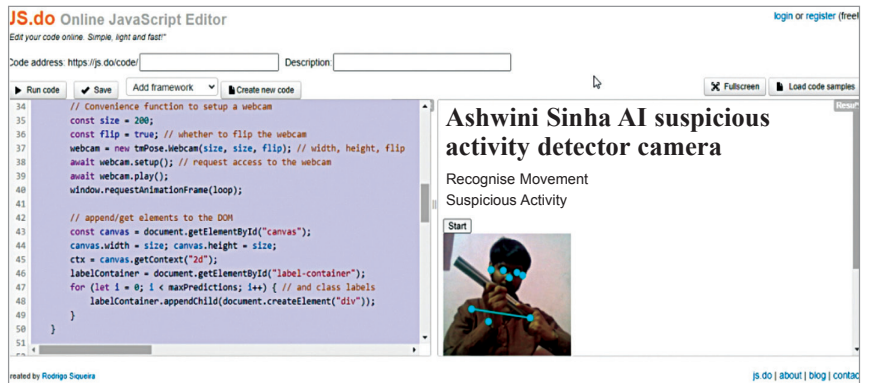


Fig. 2: Author's prototype

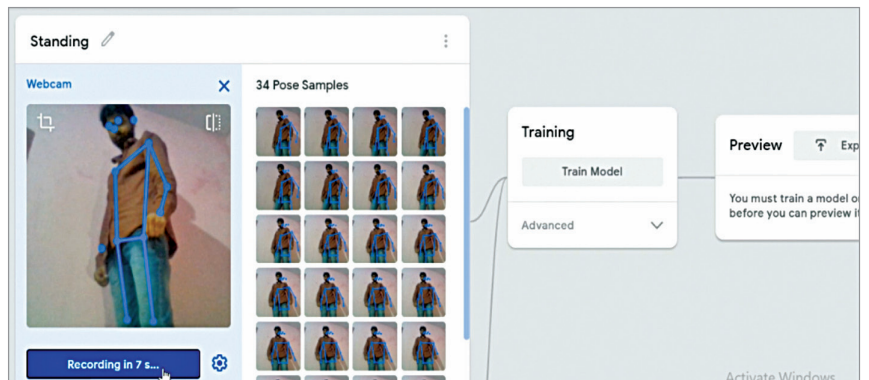


Fig. 3: Feeding the normal activity datasets to create ML model